Technical Note: Lorenz–Mie Scattering Theory

Mathematical Formulation and Computational Implementation

N. Moteki

Last updated: 2026-03-19 (corresponds to MieScat_Py v0.2.0)

Contents

1 Introduction **2**

2 Mie theory for neutral spheres **2**

2.1 Problem definition and notation 2

2.2 Ricatti–Bessel functions and logarithmic derivative 2

2.3 Partial wave coefficients 3

2.4 Optical efficiency factors 3

2.5 Scattering amplitudes 3

2.6 Mass-normalized cross sections 4

3 Extension to charged spheres **4**

4 Complex Amplitude Sensing observables **5**

5 Computational implementation **5**

1 Introduction

MieScat_Py solves the classical problem of electromagnetic scattering by a single homogeneous isotropic sphere embedded in a nonabsorbing medium. The implementation follows the formulations in Bohren and Huffman [1] (hereafter BH83) and Mishchenko et al. [2], with an extension to electrically charged spheres based on Klacka and Kocifaj [3] (hereafter KK10). The forward and backward complex scattering amplitudes used for Complex Amplitude Sensing (CAS) are defined according to Moteki [4] and Moteki and Adachi [5].

This note documents the mathematical formulations, numerical algorithms, and their correspondence to the code. Sec. 2 covers the standard Mie theory for neutral spheres, from the problem definition through to the computed observables. Sec. 3 describes the extension to charged spheres. Sec. 4 defines the CAS-specific observables. Sec. 5 summarizes the computational implementation.

2 Mie theory for neutral spheres

2.1 Problem definition and notation

Consider a sphere of diameter d_p (radius $r_\mathrm{p} = d_\mathrm{p}/2$) with complex refractive index

$$m_\mathrm{p} = n_\mathrm{p} + i\,\kappa_\mathrm{p} \tag{1}$$

embedded in a nonabsorbing medium with real refractive index m_m. A monochromatic plane wave of vacuum wavelength λ_0 illuminates the sphere. The vacuum wavenumber, medium wavenumber, size parameter, and relative refractive index are

$$k_0 = \frac{2\pi}{\lambda_0}, \qquad k = m_\mathrm{m}\,k_0, \qquad x = k\,r_\mathrm{p} = \frac{\pi\,d_\mathrm{p}\,m_\mathrm{m}}{\lambda_0}, \qquad m_\mathrm{r} = \frac{m_\mathrm{p}}{m_\mathrm{m}}. \tag{2}$$

The number of terms retained in the partial wave expansion is

$$N_\mathrm{stop} = \lfloor |x| + 4\,|x|^{1/3} + 2 \rfloor, \tag{3}$$

following the criterion of BH83.

2.2 Ricatti–Bessel functions and logarithmic derivative

The Mie solution is expressed in terms of the Ricatti–Bessel functions

$$\psi_n(z) = z\,j_n(z), \qquad \chi_n(z) = -z\,y_n(z), \qquad \xi_n(z) = \psi_n(z) + i\,\chi_n(z) = z\,h_n^{(1)}(z), \tag{4}$$

where j_n, y_n, and $h_n^{(1)}$ are the spherical Bessel, Neumann, and Hankel functions of the first kind, respectively.

The logarithmic derivative of ψ_n at argument $y = m_\mathrm{r}\,x$ is

$$D_n(y) = \frac{\mathrm{d}}{\mathrm{d}y}\ln\psi_n(y) = \frac{\psi_n'(y)}{\psi_n(y)}. \tag{5}$$

It is computed by the numerically stable *downward* recurrence (BH83, Sec. 4.8)

$$D_{n-1}(y) = \frac{n}{y} - \frac{1}{D_n(y) + n/y}, \qquad n = N_\mathrm{mx},\, N_\mathrm{mx}-1,\, \ldots,\, 1, \tag{6}$$

starting from $D_{N_\mathrm{mx}}(y) = 0$, where $N_\mathrm{mx} = \lfloor \max(N_\mathrm{stop},\, |y|) + 15 \rfloor$.

$\psi_n(x)$ is computed via the ratio $R_n = \psi_n(x)/\psi_{n-1}(x)$ using downward recurrence (Mishchenko et al. [2]):

$$R_n = \frac{1}{(2n+1)/x - R_{n+1}}, \qquad n = N_{\rm mx}-1,\,\ldots,\,0, \qquad R_{N_{\rm mx}} = \frac{x}{2N_{\rm mx}+1}, \tag{7}$$

followed by $\psi_0(x) = R_0 \cos x$ and $\psi_n(x) = R_n\, \psi_{n-1}(x)$ for $n \geq 1$.

$\chi_n(x)$ is computed by upward recurrence, which is stable for this function:

$$\chi_0(x) = -\cos x, \qquad \chi_1(x) = \frac{\chi_0(x)}{x} - \sin x, \qquad \chi_n(x) = \frac{2n-1}{x}\, \chi_{n-1}(x) - \chi_{n-2}(x). \tag{8}$$

2.3 Partial wave coefficients

For a neutral (uncharged) sphere, the partial wave (Mie) coefficients a_n and b_n are (BH83, Eq. 4.88)

$$a_n = \frac{\big(D_n(y)/m_{\rm r} + n/x\big)\, \psi_n(x) - \psi_{n-1}(x)}{\big(D_n(y)/m_{\rm r} + n/x\big)\, \xi_n(x) - \xi_{n-1}(x)}, \tag{9}$$

$$b_n = \frac{\big(m_{\rm r}\, D_n(y) + n/x\big)\, \psi_n(x) - \psi_{n-1}(x)}{\big(m_{\rm r}\, D_n(y) + n/x\big)\, \xi_n(x) - \xi_{n-1}(x)}, \tag{10}$$

where $y = m_{\rm r}\, x$ and $n = 1, 2, \ldots, N_{\rm stop}$. The coefficients a_n correspond to the electric (TM) multipoles and b_n to the magnetic (TE) multipoles.

2.4 Optical efficiency factors

Given a_n and b_n, the optical efficiency factors are (BH83, Eqs. 4.61–4.62)

$$Q_{\rm sca} = \frac{2}{x^2} \sum_{n=1}^{N_{\rm stop}} (2n+1)\big(|a_n|^2 + |b_n|^2\big), \tag{11}$$

$$Q_{\rm ext} = \frac{2}{x^2} \sum_{n=1}^{N_{\rm stop}} (2n+1)\ {\rm Re}(a_n + b_n), \tag{12}$$

$$Q_{\rm abs} = Q_{\rm ext} - Q_{\rm sca}, \tag{13}$$

$$Q_{\rm back} = \frac{1}{x^2} \left| \sum_{n=1}^{N_{\rm stop}} (2n+1)(-1)^n (a_n - b_n) \right|^2. \tag{14}$$

These are dimensionless ratios of the respective cross sections to the geometric cross section $\pi r_{\rm p}^2$.

2.5 Scattering amplitudes

The angle-dependent functions $\pi_n(\cos\theta)$ and $\tau_n(\cos\theta)$ are defined as

$$\pi_n(\cos\theta) = \frac{P_n^1(\cos\theta)}{\sin\theta}, \qquad \tau_n(\cos\theta) = \frac{{\rm d}P_n^1(\cos\theta)}{{\rm d}\theta}, \tag{15}$$

where P_n^1 is the associated Legendre function of degree n and order 1. They satisfy the recurrence relations (with $\mu = \cos\theta$, $\pi_1 = 1$, $\pi_2 = 3\mu$)

$$\pi_n = \frac{2n-1}{n-1}\, \mu\, \pi_{n-1} - \frac{n}{n-1}\, \pi_{n-2}, \qquad \tau_n = n\, \mu\, \pi_n - (n+1)\, \pi_{n-1}. \tag{16}$$

The elements of the 2×2 amplitude scattering matrix (BH83, Eq. 4.74) are

$$S_1(\theta) = \sum_{n=1}^{N_{\text{stop}}} \frac{2n+1}{n(n+1)} \big(a_n\, \pi_n + b_n\, \tau_n\big), \tag{17}$$

$$S_2(\theta) = \sum_{n=1}^{N_{\text{stop}}} \frac{2n+1}{n(n+1)} \big(a_n\, \tau_n + b_n\, \pi_n\big). \tag{18}$$

The code converts these to the amplitude matrix convention of Mishchenko et al. [2]:

$$S_{11}(\theta) = \frac{S_2(\theta)}{-ik}, \qquad S_{22}(\theta) = \frac{S_1(\theta)}{-ik}. \tag{19}$$

Note the index swap: S_{11} is derived from S_2, and S_{22} from S_1. Both have the dimension of length.

2.6 Mass-normalized cross sections

The geometric cross section, particle volume, and particle mass are

$$C_{\text{geo}} = \frac{\pi d_{\text{p}}^2}{4}, \qquad V_{\text{p}} = \frac{\pi d_{\text{p}}^3}{6}, \qquad M_{\text{p}} = \rho_{\text{p}}\, V_{\text{p}}, \tag{20}$$

where ρ_{p} is the particle material density. The mass-normalized cross sections are

$$\text{MSC} = \frac{C_{\text{geo}}\, Q_{\text{sca}}}{M_{\text{p}}}, \qquad \text{MEC} = \frac{C_{\text{geo}}\, Q_{\text{ext}}}{M_{\text{p}}}, \qquad \text{MAC} = \frac{C_{\text{geo}}\, Q_{\text{abs}}}{M_{\text{p}}}. \tag{21}$$

When d_{p} is given in m and ρ_{p} in $\text{g}\,\text{cm}^{-3}$, the code internally converts V_{p} to cm^3 so that M_{p} is in g and the mass-normalized cross sections are in $\text{m}^2\,\text{g}^{-1}$.

3 Extension to charged spheres

For a sphere carrying n_{e} elementary charges e in vacuum ($m_{\text{m}} = 1$), the formulation of KK10 introduces a charge-dependent G-parameter that modifies a_n and b_n. The total charge is $Q = n_{\text{e}}\, e$ and the electrostatic surface potential is

$$V_{\text{surf}} = \frac{Q}{4\pi\epsilon_0\, r_{\text{p}}}. \tag{22}$$

The quantum scattering rate is estimated as (KK10, Eq. 37)

$$\gamma_{\text{s}} = f_\gamma\, \frac{k_{\text{B}} T}{\hbar}, \tag{23}$$

where f_γ is a dimensionless correction factor, k_{B} is the Boltzmann constant, $T = 298\,\text{K}$, and $\hbar$ is the reduced Planck constant. The dimensionless charge parameter is (KK10, Eq. 26)

$$G = \frac{e\, V_{\text{surf}}}{m_{\text{e}}\, c^2} \cdot \frac{1}{x} \cdot \frac{1 + i\,(\gamma_{\text{s}}/\omega)}{1 + (\gamma_{\text{s}}/\omega)^2}, \tag{24}$$

where m_{e} is the electron mass, c is the speed of light, and $\omega = k\,c$ is the angular frequency.

The modified partial wave coefficients for the charged sphere are

$$a_n = \frac{\big[(1 + nG/x)\, D_n/m_{\text{p}} + n/x\big]\, \psi_n - (1 + G\, D_n/m_{\text{p}})\, \psi_{n-1}}{\big[(1 + nG/x)\, D_n/m_{\text{p}} + n/x\big]\, \xi_n - (1 + G\, D_n/m_{\text{p}})\, \xi_{n-1}}, \tag{25}$$

$$b_n = \frac{\big[m_{\text{p}}\, D_n + (n/x)(1 - Gx/n)\big]\, \psi_n - \psi_{n-1}}{\big[m_{\text{p}}\, D_n + (n/x)(1 - Gx/n)\big]\, \xi_n - \xi_{n-1}}, \tag{26}$$

where all Ricatti–Bessel functions are evaluated at x, and $D_n = D_n(m_\mathrm{p}\,x)$ with $m_\mathrm{r} = m_\mathrm{p}$ (since the medium is vacuum). When $G = 0$ (i.e., $n_\mathrm{e} = 0$), (25)–(26) reduce to the standard Mie expressions (9)–(10) with $m_\mathrm{r} = m_\mathrm{p}$. The efficiency factors, scattering amplitudes, and mass-normalized cross sections are then computed identically to Sec. 2.4–2.6.

4 Complex Amplitude Sensing observables

For the CAS protocol [4, 5], the relevant observables are the forward scattering amplitude $S(0)$ and a backward scattering amplitude S_bak.

At $\theta = 0$ the angular functions satisfy $\pi_n(1) = \tau_n(1) = 1$ for all n, so $S_1(0) = S_2(0)$ and hence $S_{11}(0) = S_{22}(0)$. The forward scattering amplitude is

$$S(0) = \frac{S_{11}(0) + S_{22}(0)}{2} = S_{11}(0). \tag{27}$$

At $\theta = \pi$ the angular functions satisfy $\pi_n(-1) = (-1)^{n+1}$ and $\tau_n(-1) = (-1)^n$, giving $S_1(\pi) = -S_2(\pi)$. The backward scattering amplitude for CAS-v2 is

$$S_\mathrm{bak} = \frac{-S_{11}(\pi) + S_{22}(\pi)}{\sqrt{2}}. \tag{28}$$

The code returns $\mathrm{Re}[S(0)]$, $\mathrm{Im}[S(0)]$, $\mathrm{Re}[S_\mathrm{bak}]$, and $\mathrm{Im}[S_\mathrm{bak}]$, all having the dimension of length.

5 Computational implementation

Table 1: Modules in `MieScat_Py` v0.2.0 and their roles.

Module	Role
`_mie_core.py`	Shared helper functions ((6)–(19))
`miescat.py`	Neutral sphere: Q-factors, S_{11}, S_{22}, MSC/MEC/MAC
`miescat_charged.py`	Charged sphere (KK10): adds V_surf, G-parameter
`mie_complex_amplitudes.py`	CAS observables: $S(0)$ and S_bak

The key numerical considerations are as follows. The logarithmic derivative $D_n(y)$ is computed by *downward* recurrence (6), which is numerically stable for complex y (BH83, Sec. 4.8). $\psi_n(x)$ is computed via the ratio method (7) using downward recurrence on the ratios R_n, which avoids the catastrophic cancellation that plagues upward recurrence for large x (Mishchenko et al. [2]). $\chi_n(x)$ is computed by upward recurrence (8), which is stable for this function. The recurrence relations for D_n, R_n, ψ_n, χ_n, π_n, and τ_n remain as sequential loops because each step depends on the previous value.

In v0.2.0, the following element-wise Python loops were replaced by NumPy vectorized operations: (i) the weight arrays $(2n+1)$, $(2n+1)/[n(n+1)]$, and $(-1)^n$; (ii) the partial wave coefficients a_n, b_n ((9)–(10) or (25)–(26)) for all n simultaneously; (iii) the scattering amplitude sums (17)–(18) over n for all angles θ simultaneously via array broadcasting.

Table 2: Correspondence between mathematical symbols and code variables.

Symbol	Code variable	Description
λ_0	`wl_0`	Vacuum wavelength
m_{m}	`m_m`	Medium refractive index
d_{p}	`d_p`	Particle diameter
m_{p}	`m_p`	Complex particle refractive index
m_{r}	`m_r`	Relative refractive index
x	`x`	Size parameter
N_{stop}	`nstop`	Number of expansion terms
$D_n(y)$	`DD[n]`	Logarithmic derivative
$\psi_n(x)$	`psi[n]`	Ricatti–Bessel (1st kind)
$\chi_n(x)$	`chi[n]`	Ricatti–Bessel (2nd kind)
$\xi_n(x)$	`xi[n]`	Ricatti–Bessel (3rd kind)
a_n	`a[n]`	Electric multipole coefficient
b_n	`b[n]`	Magnetic multipole coefficient
π_n	`pi_n[n,:]`	Angular function
τ_n	`tau_n[n,:]`	Angular function
S_1, S_2	`S1`, `S2`	BH83 amplitude elements
S_{11}, S_{22}	`S11`, `S22`	Mishchenko amplitude elements
G	`g`	Charge parameter (KK10)
V_{surf}	`surf_potential`	Surface potential
ρ_{p}	`dens_p`	Particle density

References

[1] C. F. Bohren and D. R. Huffman, *Absorption and Scattering of Light by Small Particles*, John Wiley & Sons, 1983.

[2] M. I. Mishchenko, L. D. Travis, and A. A. Lacis, *Scattering, Absorption, and Emission of Light by Small Particles*, Cambridge University Press, 2002.

[3] J. Klacka and M. Kocifaj, "On the scattering of electromagnetic waves by a charged sphere," *Progress In Electromagnetics Research*, vol. 109, pp. 17–35, 2010.

[4] N. Moteki, "Measuring the complex forward-scattering amplitude of single particles by self-reference interferometry: CAS-v1 protocol," *Optics Express*, vol. 29, no. 13, pp. 20688–20714, 2021. doi:10.1364/OE.423175.

[5] N. Moteki and K. Adachi, "Measuring the polarized complex forward-scattering amplitudes of single particles in unbounded fluid flow: CAS-v2 protocol," *Optics Express*, vol. 32, no. 21, pp. 36500–36522, 2024. doi:10.1364/OE.533776.

Acknowledgment. This document was prepared with the assistance of Claude (Anthropic). The author assumes full responsibility for the content.